Do the decks built from his games ask what they say they ask?

The misses deck's theme words and the guess-the-plan quiz, read against their own engine lines

Chess Coach research lab

Abstract

Two practice decks are built from the player's own games. The misses deck files each position where he went wrong under a theme word, which orders a session and, for most words, names the skill a solve counts toward. Reading each of its 1,336 engine-checked positions by what its own stored lines show, 37% are clear defensive misses (he had to stop something) and 21% clear missed wins. The deck files 56% of its positions under a phase name, among them 280 clear defensive misses and 144 clear missed wins, and only 40% of the clear defensive misses carry its defensive-move word. The guess-the-plan quiz has 48 positions; by the app's own rule 17 of them have two right answers out of three, the best option is a capture, a check or a two-pawn lead in 31, and seven ask a plan question with one right answer. A search with the move passed to his opponent finds a threat in 42% of the positions: 31% of those whose lines show no material change, while 41% of the clear defensive misses had no threat to answer, so his own move made the loss. The deck's 316 quiet positions hold 73 plan questions with one right answer.

1 The question

The misses deck is the player's own mistakes as practice. Each position carries a theme word from the app's labeller of the best move (`themeOfBest`): a motif when one of its detectors fires (a mate in one, a back-rank mate, a fork, a raid on f7, a free piece, a defensive move), else the phase. The word interleaves a session, and for the words the learner knows it is the skill a solve or a miss is counted toward. The guess-the-plan quiz shows a position from his games with three engine moves and asks which is the plan; an option within 45 centipawns of the best counts as right. The detector-themes project scored the labeller on puzzles, which are tactics by construction. Here it is read on the deck itself (app commit `d8b771238c`), against the deck's own engine lines.

2 Method

The rules were written in the project's README before any of these numbers was computed. What the best move did is read from its stored line (the routine's deep pass, five lines at four million nodes) by the board code alone: it *mates* (the line ends in mate or scores a mate), *wins material* (his balance, counting 1, 3, 3, 5, 9, is at least two points up at the line's end, the lower of the last two balances when it ends on a capture), or neither. What his move cost is read the same way from its own stored line, or from the move and the deck's punishing line: it *allowed mate*, *lost material* (two points or more), or neither. A miss is *defensive* when the best move wins nothing and his move allowed mate or lost material, *attacking* when the best move mates or wins and his move lost nothing, and *mixed* otherwise: 84 where both happened and 477 (36%) where the stored lines show

no material change either way, so material cannot judge them. The 64 opening deviations, which have no engine lines, are left out.

3 The misses deck

word	skill it feeds	items	defensive	attacking	mixed	fits
`defensiveMove`	`defensiveMove`	405	195	56	154	195 of 405 (48%)
`hangingPiece`	`hangingPiece`	109	13	54	42	70 of 109 (64%)
`fork`	`fork`	20	1	15	4	16 of 20 (80%)
`mateIn1`	`mateIn1`	20	0	17	3	20 of 20 (100%)
`backRankMate`	`backRankMate`	1	0	0	1	0 of 1 (0%)
`middlegame`	–	406	139	81	186	not scored
`endgame`	`conv.technique`	187	68	38	81	not scored
`opening`	–	188	73	25	90	not scored

Table 1: The deck's theme words: the learner skill each feeds, how its positions read by the stored lines, and whether the word clearly fits by the README's rule (defensive move: a defensive miss; hanging piece and fork: the best move wins or mates; mate words: it mates). Phase words name no motif and are not scored.

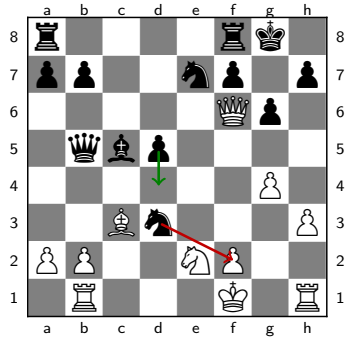
Most positions name no motif. 781 of the 1,400 positions (56%) are filed under a phase, 405 (29%) as a defensive move, and 150 (11%) under a tactic. The phase-word positions are not all formless: 280 are clear defensive misses and 144 clear missed wins. On those missed wins, the move describer the explanations use names nothing for 61%.

The defensive-move skill sees a minority of his defensive misses. Of the 489 clear defensive misses, 195 (40%) carry the defensive-move word and 294 carry another, mostly a phase. Of the 405 positions the word is on, 195 (48%) are clear defensive misses and 56 read as missed wins; the rest are mixed. The labeller calls a move defensive when one of his pieces was loose and the move protected it, which misses the threats that are not a loose piece, such as a mate.

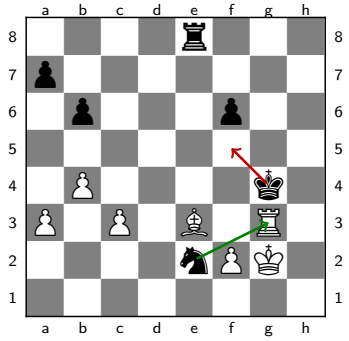
Endgame positions count toward conversion. The learner counts every position filed under `endgame` toward `conv.technique`: 187 positions, 68 of them clear defensive misses, where he was not converting anything.

The motif words. A mate in one mates in 20 of 20 positions and a fork wins or mates in 16 of 20. The free piece wins two points or more in 64% of its 109 positions; 22 more win exactly a pawn, which the two-point rule does not count, so with them it is 84%.

A mate threat filed under a phase. In `r4rk1/pp2np1p/5Qp1/1qbp4/6P1/2Bn3P/PP2NP2/1R3K1R b - - 5 17` he played 17... ♘xf2, which gave away 86 points of winning chances to 18. ♕g7#. The engine's move was 17... d4 18. ♘xd4 ♗xd4 19. ♗xd4 ♘f4+ 20. ♔g1 ♘e2+ 21. ♔h2 (`d5d4 e2d4 c5d4 c3d4 d3f4 f1g1 f4e2 g1h2`). The deck files the position under `middlegame`.



Filed under `middlegame`, feeding no skill



Filed under `endgame`, feeding `conv.technique`

Figure 1: Two positions from his games, drawn from White's side; he is the side to move. Red is his move, green the engine's.

A win filed under conversion. In `4r3/p7/1p3p2/8/1P4k1/P1P1B1R1/4nPK1/8 b - - 2 38` he played 38…♔f5 (40 points), where 38…♘xg3 39. a4 ♘e2 40. f3+ ♔f5 41. ♗d2 ♘d4 42. ♗c1 (`e2g3 a3a4 g3e2 f2f3 g4f5 e3d2 e2d4 d2c1`) wins material. The deck files it under `endgame`, so a solve counts toward `conv.technique`.

4 The guess-the-plan quiz

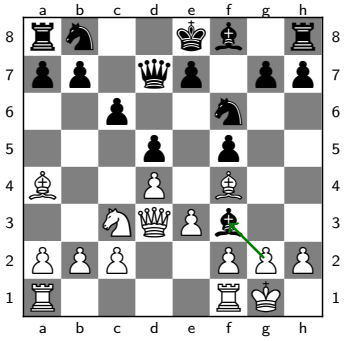


Figure 2: A quiz position: the options are 11. gxf3 (0.2), 11. ♗xb8 (−2.1), 11. ♖ae1 (−2.3), in pawns from his side. The best, a recapture, leads the next by 2.3 pawns.

By the app's rule, 31 quizzes have one right answer and 17 (35%) have two of three, where a guess is right two times in three. The best option is a capture or a check in 30 and leads the next option by two pawns or more in eleven: 31 (65%) are one or the other, a calculation question rather than a plan. In 34 he had played the best option in the game. Only seven (15%) ask a plan question with one right answer. In the position above (`rn2kb1r/pp1qp1pp/2p2n2/3p1p2/B2P1B2/2NQPb2/PPP2PPP/R4RK1 w kq - 0 11`), the best option is 11. gxf3 e6 12. e4 ♗e7 13. exf5 exf5 14. ♔h1 g6 15. ♖ae1 ♔f7 (`g2f3 e7e6 e3e4 f8e7 e4f5 e6f5 g1h1 g7g6 a1e1 e8f7`).

5 What his opponent threatened

The stored lines say what the best move and his move did, not whether he had anything to answer. So each scored position was searched by the app's engine at the routine's punishing budget (600,000 nodes), as he faced it and with the move passed to his opponent, 2,585 searches in all; the 87 positions where he was in check cannot be passed and are counted apart. By the rules the README fixed before the search ran, a position is *threatened* when the opponent's line after the pass mates or wins him two points or more, counted as the census counts a gain. 520 of the 1,249 passed positions (42%) were: 66 by mate and 454 by material. Passing cost him a median 34 points of win% in those and 14 in the rest, where it is the value of the move itself.

			threatened			the gate		
	positions	in check	n	%	mate	fires %	precision %	recall %
clear defensive miss	489	45	260	59	42	58	75	75
clear missed win	286	12	86	31	7	50	51	81
both	84	9	34	45	6	57	63	79
no material change	477	21	140	31	11	44	55	78
`defensiveMove`	405	11	271	69	39	100	69	100
`hangingPiece`	109	9	36	36	4	51	57	81
`fork`	20	0	6	30	0	60	42	83
`mateIn1`	20	0	12	60	1	80	63	83
`backRankMate`	1	0	1	100	0	100	100	100
`middlegame`	406	21	110	29	11	26	53	49
`endgame`	187	37	45	30	10	21	45	31
`opening`	188	9	39	22	1	18	52	44
all	1336	87	520	42	66	51	63	77

Table 2: The threat search by what the stored lines show and by theme word: positions, those in check (not searched), the threatened share of the searched, how many by mate, and the live coach's instant threat reader (the gate) against the search: how often it fires, how often a firing is a real threat (precision), and how many of the threats it fires on (recall).

A third of the positions with no material change were defences. Of the 456 positions whose lines show no material change, 140 (31%) were threatened: defensive misses the stored lines could not show. The other 316 are quiet: nothing was threatened, and nothing was won or lost.

Two in five clear defensive misses had no threat to answer. Of the 444 clear defensive misses searched, 260 (59%) were threatened. In the other 184 (41%) nothing was threatened before he moved: his move made the loss. Those ask what his own move allows, not what his opponent's move threatens.

The defensive-move word is right more often than the lines show. The gate fires on every defensive-move position, since the labeller's defensive move protects a piece the gate reads as loose. The search finds a threat in 271 of its 394 searched positions (69%), where the stored lines alone found a clear defensive miss in 48%. In 89 of the other 123, the loose piece was a pawn alone. The word is on 52% of all the threatened positions; phase words carry 194 (37%).

The gate misses the threats the phase words hide. Over all the searched positions the gate's precision is 63% and its recall 77%: it misses 119 threats. Of its 239 false alarms, 176 (74%) are a loose pawn alone, which the two-point rule never counts. On the 194 threatened positions filed under a phase, it fires on 44%. The rest held a threat of another kind than the three it reads (a loose piece, the f7 strike, a mate in one), and the labeller, which reads the same loose piece, could not call them defensive.

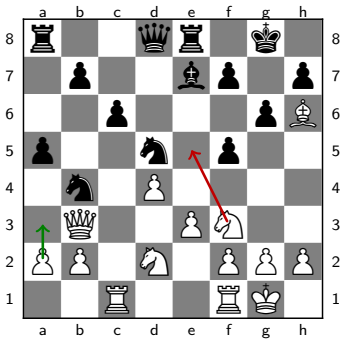


Figure 3: A quiet miss from the deck, from White's side; he is the side to move. Red is his move, green the engine's. Its three best options are 16. a3 (0.7), 16. ♖a1 (−0.1), 16. a4 (−0.4), in pawns from his side.

Quiet misses as plan questions. Of the 316 quiet positions, 73 (23%) would make a plan question with one right answer by the quiz's own rule, against the seven of the quiz's 48 positions that do today. In the first of them in the deck's order (`r2qr1k1/1p2bp1p/2p3pB/p2n1p2/1n1P4/1Q2PN2/PP1N1PPP/2R2RK1 w - - 2 16`), filed under `middlegame`, he played 16. ♘e5, which cost 16 points of winning chances; the engine's 16. a3 a4 17. ♕d1 ♘d3 18. ♖b1 ♕b6 19. b3 ♘c3 (`a2a3 a5a4 b3d1 b4d3 c1b1 d8b6 b2b3 d5c3`) is the one option within the quiz's margin.

6 What it means for the app

1. The misses deck's word should come from what the position shows before the detectors are asked. One more search per miss, with the move passed, at the budget the routine already uses to punish, tells the three kinds apart: a threat he had to answer, a loss his own move made, and a quiet position.
2. A phase word should not feed a skill: the endgame positions that were defensive misses should not count toward conversion.
3. The quiz should be drawn from positions where one option is right and the best option is quiet, or say when a position is a calculation question. The misses deck's quiet positions supply 73 such positions.
4. The live coach's instant gate misses 119 of the 520 threats the search finds, most of them filed under a phase, and a loose pawn is most of what it flags that the search does not count.

7 Limits

The rule reads material only. A capture that removes an attacker and also wins material reads as a missed win: 15 of the 34 defensive-move positions read as wins gain exactly two points, and some of those are such captures, where the word is right. The threat search reads what the opponent would do with a free move, which is not a fair question where moving at all hurts him, and it counts a threat that his best move happened to stop; the two-point rule leaves out a pawn. The lines are the engine's, at the budgets the routine uses.